

PART 2 → Fitting (0,0.5), (2,3.5),
(1,1.5), (3,7.5)

INITIAL SUMS: $\sum x = 6$, $\sum x^2 = 14$, $\sum x^3 = 36$, $\sum x^4 = 98$

$\sum y = 13$, $\sum xy = 31$, $\sum x^2y = 83$

1. SETUP → LINE, $y = mx + b$

general formula: $MSE(m, b) = \frac{1}{n} \sum_{i=1}^n (y_i - (mx_i + b))^2$

$MSE(m, b) = \frac{1}{4} [(0.5 - b)^2 + (3.5 - (2m + b))^2 + (1.5 - (m + b))^2 + (7.5 - (3m + b))^2]$

2. SETUP → PARABOLA, $y = ax^2 + bx + c$

general formula: $MSE(a, b, c) = \frac{1}{n} \sum_{i=1}^n (y_i - (ax_i^2 + bx_i + c))^2$

$MSE(a, b, c) = \frac{1}{4} [(0.5 - c)^2 + (3.5 - (4a + 2b + c))^2 + (1.5 - (a + b + c))^2 + (7.5 - (9a + 3b + c))^2]$

① $y = mx + b$

→ taken of general function first

analytical: set both partial derivatives to zero

$\frac{\partial MSE}{\partial m} = \frac{1}{n} \sum_{i=1}^n 2(y_i - (mx_i + b)) \times (-x_i)$

$= -\frac{2}{n} \sum_{i=1}^n (x_i y_i - mx_i^2 - bx_i)$

$= -\frac{2}{n} [\sum xy - m \sum x^2 - b \sum x]$

set to zero

$0 = -\frac{1}{2} [\sum xy - m \sum x^2 - b \sum x]$

$\sum xy = m \sum x^2 + b \sum x$

→ sub + solve system of equations

$14m + 6b = 31$

$6m + 4b = 13$

→ $b = \frac{13 - 6m}{4}$

$\frac{\partial MSE}{\partial b} = \frac{1}{n} \sum_{i=1}^n 2(y_i - (mx_i + b))$

$= \dots = -\frac{1}{2} [\sum y - m \sum x - bn]$

↳ similar to $\frac{\partial MSE}{\partial m}$

$\Rightarrow \sum y = m \sum x + bn$

$14m + \frac{6(13 - 6m)}{4} = 31$

$\Rightarrow m = 2.3, b = -0.2, y = 2.3x - 0.2$

MSE calculation

(0, 0.5) → $y(0) = 2.3(0) - 0.2 = -0.2$

(1, 1.5) → $y(1) = 2.3(1) - 0.2 = 2.1$

(2, 3.5) → $y(2) = 2.3(2) - 0.2 = 4.4$

(3, 7.5) → $y(3) = 2.3(3) - 0.2 = 6.7$

residual

0.7

-0.4

-0.9

0.8

→ $MSE_{line} =$

$\frac{0.49 + 0.36 + 0.81 + 0.64}{4}$

$= 0.575$

numerical: Newton-Raphson multivariate

↳ update all parameters at once using gradient + Hessian

$\theta_{n+1} = \theta_n - H^{-1} \nabla MSE(\theta_n)$

$\nabla MSE(m, b) = -\frac{2}{n} \begin{bmatrix} \sum xy - m \sum x^2 - b \sum x \\ \sum y - m \sum x - bn \end{bmatrix}$

$H = \frac{2}{n} \begin{bmatrix} \sum x^2 & \sum x \\ \sum x & n \end{bmatrix} \xrightarrow{n=4} H = \begin{bmatrix} 7 & 3 \\ 3 & 2 \end{bmatrix}$

$\det H = 5$

$H^{-1} = \frac{1}{5} \begin{bmatrix} 2 & -3 \\ 3 & 7 \end{bmatrix}$

START @ $\theta_0 = (0, 0)$: $\nabla MSE(0, 0) = \begin{bmatrix} -15.5 \\ -6.5 \end{bmatrix}$

$\theta_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \frac{1}{5} \begin{bmatrix} 2 & -3 \\ 3 & 7 \end{bmatrix} \begin{bmatrix} -15.5 \\ -6.5 \end{bmatrix} = \begin{bmatrix} 2.3 \\ -0.2 \end{bmatrix}$

$\Rightarrow (m, b) = (2.3, -0.2)$ in one step

$$\textcircled{2} y = ax^2 + bx + c$$

analytical : set partials to 0

$$MSE(a,b,c) = \frac{1}{n} \sum_{i=1}^n (y_i - (ax_i^2 + bx_i + c))^2$$

$$\begin{aligned} \frac{\partial MSE}{\partial a} &= \frac{2}{n} \sum_{i=1}^n (y_i - (ax_i^2 + bx_i + c)) \cdot (-x_i^2) \\ &= -\frac{2}{n} [\sum x_i^2 y - a \sum x_i^4 - b \sum x_i^3 - c \sum x_i^2] \end{aligned}$$

$$\Rightarrow a \sum x_i^4 + b \sum x_i^3 + c \sum x_i^2 = \sum x_i^2 y$$

$$98a + 36b + 14c = 83 \quad (1)$$

$$\frac{\partial MSE}{\partial b} = \frac{2}{n} \sum_{i=1}^n (y_i - (ax_i^2 + bx_i + c)) \cdot (-x_i)$$

$$\Rightarrow a \sum x_i^3 + b \sum x_i^2 + c n = \sum x_i y$$

$$14a + 6b + 4c = 13 \quad (2)$$

$$\frac{\partial MSE}{\partial c} = \frac{2}{n} \sum_{i=1}^n (y_i - (ax_i^2 + bx_i + c)) \cdot (-1)$$

$$a \sum x_i^2 + b \sum x_i + c \sum 1 = \sum x_i y$$

$$\Rightarrow 36a + 14b + 6c = 31 \quad (2)$$

solve system of equations, interim steps excluded:

$$a = 0.75, \quad b = 0.05 + c = 0.95$$

$$y = 0.75x^2 + 0.95x + 0.55$$

MSE calculation

$$(0, 0.5) \rightarrow y(0) = 0.55$$

residual

$$-0.05$$

$$(1, 1.5) \rightarrow y(1) = 1.35$$

$$0.15$$

$$\rightarrow MSE_{prob} = \frac{2(0.05)^2 + 2(0.15)^2}{4}$$

$$(2, 3.5) \rightarrow y(2) = 3.45$$

$$-0.15$$

$$= 0.0125$$

$$(3, 7.5) \rightarrow y(3) = 7.45$$

$$0.05$$

numerical $\rightarrow$ gradient + Hessian

*second partial derivatives from functions above

$$\rightarrow H = \frac{2}{n} \begin{bmatrix} \sum x_i^4 & \sum x_i^3 & \sum x_i^2 \\ \sum x_i^3 & \sum x_i^2 & \sum x_i \\ \sum x_i^2 & \sum x_i & n \end{bmatrix} \Rightarrow H = \begin{bmatrix} 49 & 18 & 7 \\ 18 & 7 & 3 \\ 7 & 3 & 4 \end{bmatrix}$$

$$@ \theta_0 = (0, 0, 0)$$

$$\nabla MSE(0, 0, 0) = -\frac{2}{4} \begin{bmatrix} 83 \\ 31 \\ 13 \end{bmatrix} = \begin{bmatrix} -41.5 \\ -15.5 \\ -6.5 \end{bmatrix}$$

$$\Rightarrow H\theta_1 = -\nabla MSE(0, 0, 0)$$

$$\begin{bmatrix} 49 & 18 & 7 \\ 18 & 7 & 3 \\ 7 & 3 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 41.5 \\ 15.5 \\ 6.5 \end{bmatrix}$$

*solve system, intermediate steps

$$\Rightarrow (0.10, c) = (0.75, 0.95, 0.55)$$

ONE STEP!